

STATISTICAL METHODS AND ESTIMATION

Solutions to Case Studies 1 through 5

Problem 1: E-Commerce (Customer Purchase Estimation)

1. Given Information:

- Total population size (N) = 50,000 customers
- Sample size (n) = 500 customers
- Sample proportion ($\hat{p}$) = 68% = 0.68

2. Point Estimate of Population Proportion:

The sample proportion $\hat{p}$ serves as the best point estimate for the true population proportion p :

$$\hat{p} = 0.68 \text{ (or 68\%)}$$

The estimated total number of purchasers in the entire customer base (X) is given by:

$$X = N \times \hat{p} = 50,000 \times 0.68 = 34,000 \text{ customers}$$

3. Standard Error Calculation:

Applying the finite population correction factor:

$$SE(\hat{p}) = \sqrt{[(\hat{p}(1 - \hat{p}) / n) \times ((N - n) / (N - 1))]}$$
$$SE(\hat{p}) = \sqrt{[(0.68 \times 0.32 / 500) \times (49,500 / 49,999)]} \approx 0.0207 \text{ (2.07\%)}$$

4. Conclusion:

An estimated 68% (or 34,000 customers) made a purchase during the promotional campaign. This point estimate provides the primary baseline for demand forecasting and campaign ROI evaluation.

Problem 2: Manufacturing (Component Mean Lifetime)

1. Given Information:

- Sample size (n) = 100 components
- Sample mean lifetime ($\bar{x}$) = 1,250 hours

2. Point Estimate of Population Mean:

The sample mean $\bar{x}$ is the point estimate of the true population mean lifetime μ :

$$\hat{\mu} = \bar{x} = 1,250 \text{ hours}$$

3. Estimator Properties:

- **Unbiasedness:** The expected value of the sample mean equals the true mean, $E(\bar{x}) = \mu$.
- **Consistency:** As the sample size n increases, $\bar{x}$ converges in probability to μ .
- **Efficiency:** $\bar{x}$ possesses the smallest variance among all linear unbiased estimators (BLUE).

4. Industrial Conclusion:

The point estimate for the average lifetime of all produced electronic components is 1,250 hours. This metric represents the Mean Time Between Failures (MTBF) and serves as the benchmark for establishing standard warranty terms and maintenance schedules.

Problem 3: Banking (Customer Transaction Value and Variance)

1. Given Information:

- Sample size (n) = 200 customers
- Sample mean ($\bar{x}$) = ₹18,500
- Sample standard deviation (s) = ₹4,200

2. Point Estimates:

- **Population Mean (μ):**

$$\hat{\mu} = \bar{x} = \text{₹}18,500$$

- **Population Variance (σ^2):**

$$\hat{\sigma}^2 = s^2 = (4,200)^2 = 17,640,000 \text{ ₹}^2$$

- **Standard Error of the Mean:**

$$SE(\bar{x}) = s / \sqrt{n} = 4,200 / \sqrt{200} \approx \text{₹}296.98$$

3. Financial Conclusion:

The average monthly transaction value per customer is estimated to be ₹18,500 with a variance of 17,640,000 ₹². Bessel's correction ($n - 1 = 199$) ensures that the sample variance is an unbiased estimator. These estimates assist the bank in determining daily cash reserve requirements and establishing transaction monitoring thresholds.

Problem 4: Healthcare Analytics (Adverse Side Effect Proportion)

1. Given Information:

- Total sampled patients (n) = 400 patients
- Number of patients with side effects (x) = 52 patients

2. Point Estimate of Proportion:

$$\hat{p} = x / n = 52 / 400 = 0.13 \text{ (or 13.0\%)}$$

Standard Error of the proportion:

$$SE(\hat{p}) = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[(0.13 \times 0.87) / 400]} \approx 0.0168 \text{ (1.68\%)}$$

3. Clinical Interpretation:

The point estimate of 0.13 indicates that approximately 13% of all patients in the broader clinical population receiving this treatment are expected to experience the side effect. This figure is critical for safety assessments and risk-benefit analyses in clinical documentation.

Problem 5: Food Industry (10-Step Hypothesis Testing & Confidence Interval)

Given Data: Sample size $n = 64$ packets, Sample mean $\bar{x} = 498$ g, Sample standard deviation $s = 12$ g, Confidence level = 95%.

Step	Item	Working & Details
1	Population and Sample	Population: All food packets produced by the manufacturing line. Sample: Random sample of $n = 64$ packets.
2	Parameter of Interest	True population mean packet weight (μ) in grams.
3	Point Estimate	$\hat{\mu} = \bar{x} = 498$ g
4	Null Hypothesis (H_0)	$H_0: \mu = 500$ g (The mean weight matches the nominal 500 g specification).
5	Alternative Hypothesis (H_1)	$H_1: \mu \neq 500$ g (Two-tailed test: the mean weight deviates from 500 g).
6	Significance Level (α)	$\alpha = 1 - 0.95 = 0.05$ (5%). Critical value: $z_{0.025} = \pm 1.96$.
7	Confidence Interval & Test Statistic	Standard Error: $SE = s / \sqrt{n} = 12 / \sqrt{64} = 1.5$ g Margin of Error: $ME = 1.96 \times 1.5 = 2.94$ g 95% Confidence Interval: $498 \pm 2.94 = [495.06 \text{ g}, 500.94 \text{ g}]$ Calculated Test Statistic: $z = (\bar{x} - \mu_0) / SE = (498 - 500) / 1.5 = -1.333$
8	p-Value Calculation	$P(Z \leq -1.333) \approx 0.0912$ Two-tailed p -value = $2 \times 0.0912 = 0.1824$ (exact: 0.1836).

Step	Item	Working & Details
9	Comparison of p-Value with α	Since <i>p-value</i> (0.1836) > α (0.05) and $ z = 1.333 < 1.96$, we Fail to Reject the Null Hypothesis (H_0) .
10	Industry-Oriented Conclusion	Statistical Inference: There is no statistically significant evidence at the 5% level to indicate that the average packet weight differs from 500 g. Operational Decision: The 500 g target falls within the 95% confidence interval [495.06 g, 500.94 g]. Packaging machinery does not require calibration adjustments or stoppage, though regular quality monitoring should proceed.